

Compound Units: Speed, Density & Pressure

Answer sheet · 26 marks total

Section A — Speed

A1. $150 \div 30$ (1) = **5 m/s** (1)

A2. 25×8 (1) = **200 m** (1)

A3. $90 \div 45$ (1) = **2 hours** (1)

A4. $54 \div 3.6$ = **15 m/s** (1)

Section B — Density

B1. $40 \div 20$ (1) = **2 g/cm³** (1)

B2. $54 \div 2.7$ (1) = **20 cm³** (1)

B3. 1.5×1000 (1) = **1500 kg/m³** (1)

Section C — Pressure

C1. $300 \div 0.5$ (1) = **600 Pa** (1)

C2. 4000×0.25 (1) = **1000 N** (1)

C3. A sharp knife has a much smaller contact area (1) for the same force, a smaller area produces a larger pressure (pressure = force ÷ area), so it cuts through more easily (1)

Section D — Mixed & converting

D1. 40 min = $\frac{2}{3}$ hour → speed = $12 \div (\frac{2}{3})$ = **18 km/h** (1) $18 \div 3.6$ (1) = **5 m/s** (1)

D2. $250 \div 100$ = **2.5 g/cm³** (1) $\times 1000$ = **2500 kg/m³** (1)

D3. $150 \div 0.03$ = **5000 Pa** (1) halving the area (with the same force) **doubles the pressure** to 10 000 Pa (1)

Marking note: award method marks for a correctly set-up calculation even if the final answer contains an arithmetic slip, and for correct unit-conversion working even if the final compound-unit calculation is then wrong. Missing or incorrect units on an otherwise-correct numerical answer should lose the final accuracy mark, not the method mark.